

Answer Key — Biology Grade 11

Sub-Strand 1.4: Cell Division • Lesson 10: Final Model and Return to the Phenomenon: How Does the Body Make the Billions of New Cells Needed for Growth?

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
A	C	C	D	D	C	D

USE AFTER: PREDICT PHASE — Predict: debate on 'Cells just get bigger as we eat ugali, githeri and sukuma wiki'

Q1. A baby of about 3 kg grows into a teenager of 50 kg. A student says, "Cells just get bigger as we eat ugali, githeri and sukuma wiki." Which statement best explains most of the growth?

- A. The number of cells increases because existing cells divide to make new cells**
- B. Food turns directly into new cells inside the body
- C. Cells from the food we eat are absorbed and added to the body
- D. Cells only get bigger, and their number stays the same

Correct answer: A. Growth is mainly due to an increase in cell number, as existing cells divide. Food supplies raw materials and energy for this, but it does not turn directly into cells, and cell enlargement alone cannot account for the growth.

USE AFTER: OBSERVE PHASE — Observe: Expert group 1 - Mitosis and the cell cycle

Q2. A human muscle cell has 46 chromosomes. It divides by mitosis. What is produced?

- A. Two cells, each with 23 chromosomes
- B. Four cells, each with 23 chromosomes
- C. Two cells, each with 46 chromosomes that are identical to the parent cell's**
- D. Two cells, each with 46 chromosomes but with different instructions from each other

Correct answer: C. Mitosis gives two identical daughter cells with the same chromosome number as the parent cell (46). Halving the chromosome number happens in meiosis, which makes gametes.

USE AFTER: OBSERVE PHASE — Observe: confirming that a zygote divides repeatedly, 1 becomes 2, 4, 8, 16...

Q3. A zygote (1 cell) divides by mitosis. Every cell then divides once in each round, so the pattern is 1, 2, 4, 8, 16... How many cells are there after 5 rounds of division?

- A. 25
- B. 10
- C. 32**
- D. 16

Correct answer: C. The number of cells doubles in each round. After 5 rounds there are $2 \times 2 \times 2 \times 2 \times 2 = 32$ cells. Repeated doubling is how one cell can become trillions.

USE AFTER: OBSERVE PHASE — Observe: Expert group 3 - Budding in yeast and binary fission in bacteria

Q4. Yeast cells in mandazi dough and bacteria in milk both reproduce by dividing. Which statement correctly distinguishes budding from binary fission?

- A. Budding splits a cell into two equal halves; binary fission forms a small outgrowth on the parent cell
- B. Budding happens only in bacteria and binary fission happens only in yeast
- C. Both budding and binary fission make four cells with half the chromosomes
- D. Budding is an unequal division in which a small bud grows from a yeast cell and breaks off; binary fission splits a bacterium into two equal cells**

Correct answer: D. In budding, a smaller daughter cell (the bud) forms on the parent and separates, so the division is unequal. In binary fission, the cell splits into two equal parts. The other options swap or confuse these processes.

USE AFTER: EXPLAIN PHASE — Explain: Discussion diamond - 'DNA is copied before division' and the CER Final Explanation

Q5. Before a body cell divides by mitosis, its DNA is copied during interphase. Why is this important?

- A. So that food can be changed into new cells
- B. So that each daughter cell receives only half of the instructions
- C. So that the cell can become bigger before dividing
- D. So that each of the two daughter cells receives a full set of identical instructions**

Correct answer: D. Copying DNA means each daughter cell receives a complete, identical set of chromosomes. This is how every new cell gets the right instructions.

USE AFTER: DQB PHASE — Driving Question Board: sorting the question 'Can plants like bananas be grown from one cell?'

Q6. Banana plantlets are grown in a laboratory from one small piece of banana tissue. Which idea from cell division does this show?

- A. Plants can only form from the joining of two gametes
- B. The piece of tissue grows bigger but its cells never divide
- C. Cells from one piece of tissue can divide repeatedly by mitosis and form a whole new plant**
- D. Meiosis in the tissue makes many new plantlets

Correct answer: C. Tissue culture shows that a single cell or piece of tissue can divide again and again by mitosis, and its cells can specialise to form a whole organism.

USE AFTER: MODEL BUILDING PHASE — Model: gallery walk and Model Checklist (zygote from gametes by meiosis; mitosis for growth)

Q7. A group's final model of how a child grows says, "Meiosis makes the new muscle and bone cells needed for growth." What correction should the group make?

- A. Meiosis is correct, because it makes four identical body cells
- B. Growth happens only by cells getting bigger, so meiosis and mitosis should be removed
- C. Growth happens by binary fission, which makes the new muscle and bone cells
- D. Growth happens by mitosis; meiosis makes gametes with half the chromosome number**

Correct answer: D. Body cells for growth are made by mitosis, which gives identical daughter cells. Meiosis makes gametes (sperm and egg) with half the chromosomes. These join at fertilisation to form the zygote.